

Chapter 4 Assignment — Exponential and Logarithmic Functions: Strategy, Practice, and Challenges

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example* to model thinking. Then try the *Practice* (skill building) and the *Challenge* (deeper) items.

Give exact answers when possible and include domain restrictions.

Convention: $\ln$ denotes the natural logarithm. When no base is shown, $\log$ means natural log; $\log_b$ denotes base- b .

1. Topic 1: Exponential and Logarithmic Functions as Inverses

Strategy Notes. The functions $y = a^x$ (with $a > 0$, $a \neq 1$) and $y = \log_a x$ are inverses: reflect across $y = x$, swap domain/range, intercepts swap to $(1, 0)$ and $(0, 1)$ when $a = e$.

Worked Example. Describe the transformations from $y = \ln x$ to $y = \ln(x - 2) + 3$ and sketch.

Horizontal shift right 2, vertical shift up 3, vertical asymptote at $x = 2$, domain $(2, \infty)$, range $(-\infty, \infty)$.

Practice.

- (a) On the same axes, sketch $y = 3^x$ and $y = \log_3 x$. Label intercepts and asymptotes; indicate two points on each curve.
- (b) Describe the sequence of transformations taking $y = e^x$ to $y = 2e^{-x+1} - 3$. Identify horizontal asymptote and y -intercept.

Challenge.

- (a) Find a base $a > 1$ such that the graphs of $y = a^x$ and $y = \log_a x$ cross at a point with integer coordinates. Justify.

2. Topic 2: Laws of Logarithms — Expand and Condense

Strategy Notes. For $M, N > 0$: $\log_b(MN) = \log_b M + \log_b N$, $\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$, $\log_b(M^k) = k \log_b M$. Track domains before simplifying.

Worked Example. Expand $\log\left(\sqrt{\frac{x^2(x-1)}{(x+3)^5}}\right)$.

Result: $\frac{1}{2}[2\ln x + \ln(x-1) - 5\ln(x+3)]$ with domain $(-\infty, -3) \cup (1, \infty)$.

Practice.

(a) Expand completely and state domain restrictions: $\log_2\left(\frac{3x\sqrt{x-4}}{(x+1)^2}\right)$.

(b) Condense to a single logarithm: $2\ln(x+2) - \frac{1}{2}\ln(9-x^2) - \ln x$.

Challenge.

(a) Prove $\log_b\left(\frac{1}{a}\right) = -\log_b a$ for $a, b > 0$, $b \neq 1$. Then evaluate $\log_{\sqrt{2}}\left(\frac{1}{8}\right)$.

3. Topic 3: Solving Exponential and Logarithmic Equations

Strategy Notes. Isolate the exponential/log term when possible; use properties to combine logs; apply inverse functions carefully and check domains for extraneous solutions.

Worked Example. Solve $\ln(2x - 1) = 3 - \ln(x + 1)$.

Combine: $\ln\left(\frac{2x - 1}{x + 1}\right) = 3$. Exponentiate: $\frac{2x - 1}{x + 1} = e^3$. Solve for x ensuring $x > \frac{1}{2}$ and $x > -1$.

Practice.

(a) Solve $3^{x+2} = 7$; give an exact logarithmic form and a decimal approximation.

(b) Solve $\log_5(x + 4) + \log_5(x - 1) = 2$ with domain justification.

(c) Solve $e^{2x} - e^x - 6 = 0$ by substitution.

4. Topic 4: Modeling with Exponential and Logarithmic Functions

Strategy Notes. Growth/decay: $P(t) = P_0a^t$ for discrete compounding or $P(t) = P_0e^{kt}$ for continuous. Doubling time (for $k > 0$): $T_2 = \frac{\ln 2}{k}$. Half-life (for $k < 0$): $t_{1/2} = \frac{\ln 2}{|k|} = -\frac{\ln 2}{k}$.

Worked Example. A substance has half-life 5 years. If 100 g initially, find k and the time until 20 g remain.

$$k = -\frac{\ln 2}{5} \text{ and } t = \frac{\ln(20/100)}{k} \approx 11.61 \text{ years.}$$

Practice.

- (a) A culture starts at 800 and grows to 4800 in 2 hours with continuous growth. Find k , $P(t)$, and $P(3)$.
- (b) \$10,000 is invested at 6% interest compounded monthly. Find the balance after 4 years and the effective annual rate.

5. Topic 5: Domain and Range of Logarithmic Functions

Strategy Notes. For $f(x) = \log_a(g(x))$, require $g(x) > 0$. For ranges, analyze transformations of the parent function.

Worked Example. Find the domain and range of $f(x) = 3 \log_2(x^2 - 2x - 8) + 1$.

Domain: $x^2 - 2x - 8 > 0 \Rightarrow x < -2$ or $x > 4$. Range: $(-\infty, \infty)$.

Practice.

(a) Determine the domain and range of $f(x) = \ln(x^2 - 6x + 10)$.

(b) Determine the domain and range of $g(x) = \log_2(5 - \frac{1}{x})$.

Challenge.

(a) Find all real k such that $h(x) = \log_3(x^2 - kx - 6)$ has domain consisting of exactly two disjoint intervals. Describe the domain.